

Master Hardware & Timing Constraints Guide

SDC · UPF (IEEE 1801) · SpyGlass SGDC · DFT Protocols · PnR Rules · FPGA XDC/QSF

Curriculum Scope: Complete compilation of every hardware design constraint, timing assertion, environmental rule, power intent command, CDC directive, and physical boundary rule taught across the entire Digital Design Diploma.

Tools Covered: Synopsys Design Compiler (DC), PrimeTime (PT), Formality (LEC), TetraMAX (DFT), SpyGlass Lint/CDC, Cadence Innovus / Synopsys ICC2 (PnR), ModelSim/VCS (GLS), AMD/Xilinx Vivado (XDC), Intel Quartus Prime (SDC/QSF).

Constraint Domain	Format / Standard	Standard	Primary Purpose & Scope
Timing & Optimization	Synopsys (TCL)	SDC	Clocks, generated clocks, I/O delays, DRVs, exceptions
Power Intent	IEEE 1801 UPF / CPF	UPF / CPF	Power domains, power switches, level shifters, isolation, PST
CDC & Lint Rules	SpyGlass SGDC		Domain tags, asynchronous resets, quasi-static buses, waivers
Test & Scan (DFT)	SDC / SPF / STIL		Scan clocks, scan enable, chain configs, autofix rules
Signoff STA	PrimeTime TCL	SDC / TCL	OCV derates, AOCV/POCV tables, CPPR, SI crosstalk checks
Physical Design (PnR)	DEF / LEF	Tcl / Tech	Core utilization, halos, CTS options, NDR 2W2S, antenna rules
FPGA Constraints	Xilinx XDC / QSF	Intel	Physical pin mapping, I/O standards, BUFG clock placement

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1 Constraint Fundamentals & Timing Path Taxonomy

In digital ASIC and FPGA design, **Constraints** represent the mathematical contract between the hardware designer, the EDA synthesis engine, the Static Timing Analysis (STA) signoff tool, and the Physical Place-and-Route (PnR) router.

The 4 Fundamental Timing Paths

Every synchronous digital design is partitioned into four fundamental timing path categories:

- Input-to-Register (In-to-Reg):** Constrained by `set_input_delay` relative to an input clock.
- Register-to-Register (Reg-to-Reg):** Constrained by `create_clock` and `create_generated_clock`.
- Register-to-Output (Reg-to-Out):** Constrained by `set_output_delay` relative to an output clock.
- Input-to-Output (In-to-Out Feedthrough):** Constrained by both `set_input_delay` and `set_output_delay` (or `set_max_delay`).

Path Category	Startpoint	Endpoint
Reg-to-Reg	Clock pin of Launch Flip-Flop	Data input (D) pin of Capture Flip-Flop
In-to-Reg	Primary Input Port of chip	Data input (D) pin of Capture Flip-Flop
Reg-to-Out	Clock pin of Launch Flip-Flop	Primary Output Port of chip
In-to-Out	Primary Input Port of chip	Primary Output Port of chip

Table 1: Static Timing Analysis Path Definitions

2 Clock Definition & Clock Tree Constraints (SDC)

2.1 create_clock

`create_clock` [Primary Clock Specification]

Syntax:

```
create_clock -name <clk_name> -period <period_val>
             [-waveform {<rise_time> <fall_time>}]
             [<port_or_pin_list>]
```

Examples:

```
# 100 MHz Master System Clock (50% duty cycle: rise=0ns, fall=5ns)
create_clock -name SYS_CLK -period 10.0 [get_ports clk_in]

# 50 MHz Clock with asymmetric duty cycle (rise=0ns, fall=6ns)
create_clock -name ASYM_CLK -period 20.0 -waveform {0 6.0} [get_ports clk_asym
]

# Virtual Clock (Reference clock for I/O with no physical port)
create_clock -name VCLK_IN -period 10.0
create_clock -name VCLK_OUT -period 10.0
```

Physical & Timing Rationale

Physical Rationale: Clocks are the fundamental heartbeat of synchronous circuits. Defining a clock sets the target clock period T_{clk} for setup checks and establishes the time zero reference for all downstream

register-to-register paths. If no port/pin is specified, the clock is treated as a **Virtual Clock**, used strictly as a timing reference for input and output board-level delays.

2.2 create_generated_clock

create_generated_clock

[Divided / Multiplied / Inverted Clocks]

Syntax:

```
create_generated_clock -name <gen_clk_name>
                      -source <master_pin_or_port>
                      [-divide_by <int>] [-multiply_by <int>]
                      [-invert] [-duty_cycle <percent>]
                      [-edges {<edge1> <edge2> <edge3>}]
                      [-edge_shift {<shift1> <shift2> <shift3>}]
                      [-combinational]
                      <target_pin_or_port>
```

Examples:

```
# Clock Divided by 2 on Flip-Flop Q pin
create_generated_clock -name CLK_DIV2 -source [get_ports clk] \
                      -divide_by 2 [get_pins u_div2/q_reg/Q]

# Clock Divided by 4 with Phase Inversion
create_generated_clock -name CLK_DIV4_INV -source [get_ports clk] \
                      -divide_by 4 -invert [get_pins u_div4/q_reg/Q]

# Master UART TX Clock derived from Clock Divider MUX output
create_generated_clock -name TX_CLK -source [get_ports clk] \
                      -divide_by 8 [get_pins u_clkdiv_mux/clk_out]

# Pure Combinational Gated Clock (Zero frequency alteration)
create_generated_clock -name GATED_ALU_CLK -source [get_ports clk] \
                      -combinational [get_pins u_icg/CLK_OUT]
```

Physical & Timing Rationale

Physical Rationale: Generated clocks maintain phase, latency, and jitter relationships with their master source. Defining them using `create_clock` instead of `create_generated_clock` would erase source latency tracking and break clock tree balancing in CTS and STA.

2.3 set_clock_uncertainty

set_clock_uncertainty

[Jitter, Skew, and Timing Margins]

Syntax:

```
set_clock_uncertainty [-setup | -hold] <uncertainty_val>
                      [-from <clk1>] [-to <clk2>]
                      [<clock_list>]
```

Examples:

```
# Pre-CTS Setup Uncertainty (Jitter + Estimated Skew + Margin)
set_clock_uncertainty -setup 0.25 [get_clocks SYS_CLK]

# Pre-CTS Hold Uncertainty (Estimated Skew)
set_clock_uncertainty -hold 0.10 [get_clocks SYS_CLK]

# Post-CTS Clock Uncertainty (Jitter only, since skew is measured directly)
set_clock_uncertainty -setup 0.05 [get_clocks SYS_CLK]
set_clock_uncertainty -hold 0.02 [get_clocks SYS_CLK]
```

```
# Inter-Clock Domain Uncertainty
set_clock_uncertainty -setup 0.35 -from [get_clocks CLK_A] -to [get_clocks
    CLK_B]
```

Physical & Timing Rationale

Pre-CTS vs Post-CTS Difference:

- **Pre-CTS (Synthesis):** Clocks are ideal (zero latency). Uncertainty must cover PLL Jitter (T_{jitter}), estimated clock tree skew (T_{skew}), and signoff guardband margins (T_{margin}).
- **Post-CTS (PnR / Signoff):** Clocks are propagated. Physical skew is extracted directly from the routed clock tree. Uncertainty is reduced to strictly **PLL Jitter** and power supply noise margins.

2.4 set_clock_latency & set_clock_transition

set_clock_latency & set_clock_transition

[Insertion Delay and Slew]

Examples:

```
# Source Latency (Off-chip board/package clock delay)
set_clock_latency -source -early 1.2 [get_clocks SYS_CLK]
set_clock_latency -source -late 1.8 [get_clocks SYS_CLK]

# Network Latency (Pre-CTS estimated on-chip insertion delay)
set_clock_latency 0.8 [get_clocks SYS_CLK]

# Clock Transition (Clock slew rate on clock pins)
set_clock_transition 0.12 [get_clocks SYS_CLK]

# Post-CTS: Activate real propagated clock delay calculations
set_propagated_clock [all_clocks]
```

3 I/O Interface & Boundary Constraints (SDC)

3.1 set_input_delay

set_input_delay

[Input Arrival Time Constraints]

Syntax:

```
set_input_delay -max <max_delay> -clock <clk_name> [-clock_fall] [-add_delay]
    <port_list>
set_input_delay -min <min_delay> -clock <clk_name> [-clock_fall] [-add_delay]
    <port_list>
```

Examples:

```
# Max Input Delay (for Setup check): 25% of clock period (2.5ns on 10ns clock)
set_input_delay -max 2.5 -clock SYS_CLK [get_ports -filter "direction == in &&
    name != clk"]

# Min Input Delay (for Hold check): 5% of clock period (0.5ns)
set_input_delay -min 0.5 -clock SYS_CLK [get_ports -filter "direction == in &&
    name != clk"]

# DDR (Double Data Rate) Input Delay (Both Clock Edges)
set_input_delay -max 1.2 -clock DDR_CLK [get_ports ddr_data_in]
set_input_delay -max 1.2 -clock DDR_CLK -clock_fall -add_delay [get_ports
    ddr_data_in]
set_input_delay -min 0.3 -clock DDR_CLK [get_ports ddr_data_in]
set_input_delay -min 0.3 -clock DDR_CLK -clock_fall -add_delay [get_ports
```

```
ddr_data_in]
```

Input Delay Formulations

$$T_{input_delay(max)} = T_{board_clk_delay} + T_{ext_launch_cq(max)} + T_{ext_pcb_trace(max)} \quad (1)$$

$$T_{input_delay(min)} = T_{board_clk_delay} + T_{ext_launch_cq(min)} + T_{ext_pcb_trace(min)} \quad (2)$$

The allowable internal chip combinational delay T_{comb_in} is bounded by:

$$T_{comb_in} \leq T_{clk} - T_{input_delay(max)} - T_{setup} \quad (3)$$

3.2 set_output_delay

set_output_delay

[Downstream Capture Constraints]

Syntax:

```
set_output_delay -max <max_delay> -clock <clk_name> [-clock_fall] [-add_delay]
  <port_list>
set_output_delay -min <min_delay> -clock <clk_name> [-clock_fall] [-add_delay]
  <port_list>
```

Examples:

```
# Max Output Delay (External PCB trace + External Setup time): 3.0ns
set_output_delay -max 3.0 -clock SYS_CLK [get_ports -filter "direction == out"
]

# Min Output Delay (Negative of external hold time requirement): -0.5ns
set_output_delay -min -0.5 -clock SYS_CLK [get_ports -filter "direction == out"
"]
```

Output Delay Formulations

$$T_{output_delay(max)} = T_{ext_pcb_trace(max)} + T_{ext_capture_setup} \quad (4)$$

$$T_{output_delay(min)} = T_{ext_pcb_trace(min)} - T_{ext_capture_hold} \quad (5)$$

The allowable internal chip clock-to-output delay $T_{cq} + T_{comb_out}$ is bounded by:

$$T_{cq} + T_{comb_out} \leq T_{clk} - T_{output_delay(max)} \quad (6)$$

3.3 set_driving_cell & set_load

set_driving_cell & set_load

[I/O Electrical Modeling]

Examples:

```
# Model input port drive strength using an inverter from the standard cell
library
set_driving_cell -lib_cell INV_X4 -library
  scmetro_tsmc_cl013g_rvt_ss_1p08v_125c [all_inputs]

# Exclude clock and reset ports from standard cell drive model (ideal nets
pre-CTS)
set_dont_touch_network [get_ports {clk rst_n}]

# Explicit input transition time (slew) if driving cell is not used
```

```
set_input_transition 0.15 [get_ports uart_rx]

# Output capacitive load (pF or fF depending on technology library unit)
set_load 0.05 [all_outputs] ;# 50 fF capacitive load on all output
pins
set_load 5.0 [get_ports pad_tx_out] ;# 5.0 pF heavy off-chip PCB pad load
```

4 Timing Exceptions & Path Modifiers (SDC)

4.1 set_false_path

set_false_path

[Disabling Non-Functional Timing Checks]

Examples:

```
# Cut asynchronous reset deassertion path (handled via recovery/removal checks)
set_false_path -from [get_ports rst_n]

# Cut static/quasi-static configuration register paths
set_false_path -from [get_cells u_regfile/cfg_baud_div_reg*]

# Cut paths crossing asynchronous clock domains (Alternative to
set_clock_groups)
set_false_path -from [get_clocks SYS_CLK] -to [get_clocks UART_TX_CLK]

# Cut test/scan-mode paths during functional timing analysis
set_false_path -through [get_pins u_dft_mux/test_mode]
```

Physical & Timing Rationale

CRITICAL WARNING: `set_false_path` completely disables both Setup AND Hold analysis. Never apply a false path to a synchronous multicycle path or a CDC path that requires skew bounding (`set_max_delay -datapath_only`).

4.2 set_multicycle_path

set_multicycle_path

[Multi-Cycle Synchronous Datapaths]

Examples:

```
# 2-Cycle Multicycle Path (Setup takes 2 clock cycles)
set_multicycle_path 2 -setup -from [get_cells u_alu/op_reg*] -to [get_cells
u_alu/result_reg*]
set_multicycle_path 1 -hold -from [get_cells u_alu/op_reg*] -to [get_cells
u_alu/result_reg*]

# 4-Cycle Multicycle Path for Deep Pipelined Divider
set_multicycle_path 4 -setup -from [get_cells u_div/a_reg*] -to [get_cells
u_div/out_reg*]
set_multicycle_path 3 -hold -from [get_cells u_div/a_reg*] -to [get_cells
u_div/out_reg*]
```

Multicycle Hold Rule

In SDC, setting `-setup N` automatically shifts the default hold check to cycle $N - 1$. For standard synchronous single-clock designs where data must be held relative to the original launch edge (cycle 0), you **MUST** specify `-hold (N - 1)`.

4.3 set_clock_groups

set_clock_groups

[Asynchronous & Exclusive Clock Domains]

Examples:

```
# Asynchronous Clocks (System Clock vs UART TX Clock)
set_clock_groups -asynchronous \
    -group [get_clocks SYS_CLK] \
    -group [get_clocks TX_CLK]

# Logically Exclusive Clocks (Multiplexed Clock - Functional vs Fast Test Clock)
set_clock_groups -logically_exclusive \
    -group [get_clocks FUNC_CLK] \
    -group [get_clocks TEST_CLK]

# Physically Exclusive Clocks (Shared pad driven by two external mutually exclusive sources)
set_clock_groups -physically_exclusive \
    -group [get_clocks XTAL_CLK] \
    -group [get_clocks EXT_PLL_CLK]
```

4.4 set_max_delay & set_min_delay (-datapath_only)**set_max_delay -datapath_only**

[Asynchronous CDC Skew Bounding]

Examples:

```
# Constrain pure datapath latency across asynchronous FIFO Gray pointers
set_max_delay 4.0 -datapath_only -from [get_cells u_fifo/wr_ptr_gray_reg*] \
    -to [get_cells u_fifo/u_sync_wr/sync_reg1*]

# Pure Point-to-Point Min/Max Delays (No clock reference)
set_max_delay 8.0 -from [get_ports ext_req] -to [get_ports ext_ack]
set_min_delay 1.0 -from [get_ports ext_req] -to [get_ports ext_ack]
```

4.5 set_case_analysis & set_disable_timing**set_case_analysis & set_disable_timing**

[Mode Selection & Loop Breaking]

Examples:

```
# Statically configure Functional Mode (Test Mode = 0)
set_case_analysis 0 [get_ports test_mode]
set_case_analysis 0 [get_ports scan_enable]

# Configure Clock Divider MUX select pin to Slow Division Mode
set_case_analysis 1 [get_pins u_clkdiv_mux/sel_div4]

# Disable timing arc across an Integrated Clock Gating cell to break false loop
set_disable_timing -from E -to Q [get_cells u_icg]
```

5 Environmental, Design Rule (DRC) & Optimization Constraints**5.1 Design Rule Constraints (DRCs)****set_max_transition, set_max_capacitance, set_max_fanout**

[Electrical Integrity]

Examples:

```
# Max transition (slew limit) across all nodes in the design
```

```
set_max_transition 0.20 [current_design]

# Max pin capacitance across design (overriding library default)
set_max_capacitance 0.15 [current_design]

# Max fanout guideline
set_max_fanout 16 [current_design]

# Area constraint (0 forces synthesis to minimize area as much as possible
# without violating timing)
set_max_area 0
```

5.2 Path Grouping for Optimization

group_path

[Custom Optimization Weighting]

Examples:

```
# Group In-to-Reg, Reg-to-Out, and In-to-Out paths
group_path -name INREG -from [all_inputs]
group_path -name REGOUT -to [all_outputs]
group_path -name INOUT -from [all_inputs] -to [all_outputs]

# Dedicated group for Master System Clock with critical range and higher
# weight
group_path -name SYS_CLK_GRP -critical_range 1.0 -weight 2.0 [get_clocks
  SYS_CLK]
```

5.3 Operating Conditions & Wire Load Models

set_operating_conditions & set_wire_load_model

[PVT & Interconnect Modeling]

Examples:

```
# Set Slow PVT corner for pre-layout synthesis (SS, 1.08V, 125C)
set_operating_conditions -max scmetro_tsmc_cl013g_rvt_ss_1p08v_125c \
  -max_library scmetro_tsmc_cl013g_rvt_ss_1p08v_125c

# Statistical Wire Load Model (for legacy pre-layout nodes > 65nm)
set_wire_load_model -name "tsmc13_wl10" -library
  scmetro_tsmc_cl013g_rvt_ss_1p08v_125c
set_wire_load_mode enclosed
```

6 Static Timing Analysis (STA) Signoff Constraints (PrimeTime)

6.1 On-Chip Variation (OCV) Derating

set_timing_derate

[OCV Delay Margins]

Examples:

```
# Setup Timing Derate (Slow Corner: Max late launch, Min early capture)
set_timing_derate -early 0.95 -clock [current_design] ;# 5% faster capture
  clock
set_timing_derate -late 1.05 -data [current_design] ;# 5% slower launch
  datapath
set_timing_derate -late 1.05 -clock [current_design] ;# 5% slower launch
  clock

# Hold Timing Derate (Fast Corner: Min early launch, Max late capture)
set_timing_derate -early 0.90 -data [current_design] ;# 10% faster launch
```

```

data
set_timing_derate -early 0.90 -clock [current_design]  ;# 10% faster launch
clock
set_timing_derate -late 1.10 -clock [current_design]  ;# 10% slower capture
clock

# Net-specific vs Cell-specific derates
set_timing_derate -cell_delay -early 0.92 [current_design]
set_timing_derate -net_delay -late 1.08 [current_design]

```

6.2 Signoff STA Audit Script

check_timing & report_timing

[Signoff Validation Commands]

Examples:

```

# Comprehensive unconstrained endpoints audit
check_timing -include {unconstrained_endpoints no_clock loop no_input_delay
no_output_delay}

# Worst Negative Slack (WNS) Setup Timing Report
report_timing -delay_type max -max_paths 50 -path_type full_clock -derate
-crosstalk_delta > reports/setup_timing.rpt

# Hold Timing Report
report_timing -delay_type min -max_paths 50 -path_type full_clock -derate >
reports/hold_timing.rpt

# Summary constraint report
report_constraint -all_violators > reports/all_violators.rpt

```

7 Low Power Intent Constraints (UPF / IEEE 1801)

Unified Power Format (UPF)

[Power Domain, Switch, Isolation & PST]

Syntax & Master Example:

```

# 1. Define Power Domains
create_power_domain PD_TOP -include_scope
create_power_domain PD_ALU -elements {u_alu}
create_power_domain PD_MEM -elements {u_regfile}

# 2. Define Supply Ports & Supply Nets
create_supply_port VDD -direction in
create_supply_port VDD_LOW -direction in
create_supply_port VSS -direction in

create_supply_net VDD_TOP -domain PD_TOP
create_supply_net VDD_ALU -domain PD_ALU
create_supply_net VDD_ALU_SW -domain PD_ALU ;# Switched supply
create_supply_net VSS -domain PD_TOP

# 3. Connect Supply Nets to Ports
connect_supply_net VDD_TOP -ports {VDD}
connect_supply_net VSS -ports {VSS}

# 4. Set Domain Primary Power Nets
set_domain_supply_net PD_TOP -primary_power_net VDD_TOP -primary_ground_net
VSS
set_domain_supply_net PD_ALU -primary_power_net VDD_ALU_SW -primary_ground_net
VSS

```

```

# 5. Create Power Switch (Header PMOS Switch for Power Gating)
create_power_switch pwr_sw_alu \
    -domain PD_ALU \
    -input_supply_port {in_pwr VDD_TOP} \
    -output_supply_port {out_pwr VDD_ALU_SW} \
    -control_port {sleep u_pwr_ctrl/alu_sleep} \
    -on_state {alu_on in_pwr {!sleep}}

# 6. Set Isolation Strategy (Clamp to 0 when PD_ALU is powered down)
set_isolation iso_alu_out \
    -domain PD_ALU \
    -isolation_supply_set {isolation_power VDD_TOP isolation_ground VSS} \
    -clamp_value 0 \
    -applies_to outputs

set_isolation_control iso_alu_out \
    -domain PD_ALU \
    -isolation_signal u_pwr_ctrl/alu_iso_en \
    -isolation_sense high \
    -location parent

# 7. Set Level Shifter Strategy (Low-to-High: 0.8V to 1.2V)
set_level_shifter ls_low_to_high \
    -domain PD_ALU \
    -applies_to inputs \
    -rule low_to_high \
    -location self

# 8. Set State Retention Strategy
set_retention ret_alu \
    -domain PD_ALU \
    -retention_power_net VDD_TOP \
    -retention_ground_net VSS

set_retention_control ret_alu \
    -domain PD_ALU \
    -save_signal {u_pwr_ctrl/save_en high} \
    -restore_signal {u_pwr_ctrl/restore_en high}

# 9. Power State Table (PST) Definition
create_pst sys_pst -supplies {VDD_TOP VDD_ALU_SW VSS}
add_pst_state FULL_ON -pst sys_pst -state {1.20 1.20 0.0}
add_pst_state ALU_OFF -pst sys_pst -state {1.20 OFF 0.0}
add_pst_state STANDBY -pst sys_pst -state {0.80 OFF 0.0}

```

8 Clock Domain Crossing (CDC) Constraints (SpyGlass SGDC)

SpyGlass SGDC Constraints

[CDC Clocks, Resets, and Waivers]

Master Example:

```

# 1. Set Top-Level Design Scope
current_design "System_Top"

# 2. Define Clocks and Domain Tags
clock -name "clk" -domain DOMAIN_SYS -tag SYS_CLK -period 10.0
    -edge {0 5.0}
clock -name "u_uart/tx_clk" -domain DOMAIN_UART -tag UART_CLK -period 80.0
    -edge {0 40.0}

# 3. Define Asynchronous Resets
reset -name "rst_n" -value 0 -async

```

```
# 4. Declare Quasi-Static Signals (Bypasses CDC synchronizer checks)
quasi_static -name "u_regfile/baud_rate_div*"
quasi_static -name "u_regfile/config_parity_en"

# 5. Define Synchronizer Cells & Structural FIFO directives
sync_cell -name "DATA_SYNC" -input "unsync_in" -output "sync_out"
fifo -memory "u_async_fifo/mem" -rd_clk "clk" -wr_clk "u_uart/tx_clk"

# 6. Waivers for Specific Approved Architectures
waive -rule Ac_conv01 -cell "u_sys_ctrl" -comment "Controlled FSM handshake
re-convergence"
waive -rule W188 -cell "u_rom" -comment "Intentionally incomplete
ROM table"
```

9 Design for Testability (DFT) Constraints & Protocols

Synopsys DFT Compiler SDC / Tcl Constraints

[Scan Chain Insertion]

Examples:

```
# 1. Define Scan Signals
set_dft_signal -view spec -type ScanClock -port [get_ports clk]
               -active_state 1
set_dft_signal -view spec -type Reset -port [get_ports rst_n]
               -active_state 0
set_dft_signal -view spec -type ScanEnable -port [get_ports scan_en]
               -active_state 1
set_dft_signal -view spec -type ScanDataIn -port [get_ports scan_in]
set_dft_signal -view spec -type ScanDataOut -port [get_ports scan_out]

# 2. Configure Scan Architecture
set_scan_configuration -chain_count 4 \
                      -style multiplexed_flip_flop \
                      -clock_mixing mix_clocks \
                      -add_lockup true

# 3. Auto-fix Violating Non-Controllable Asynchronous Pins
set_autofix_configuration -type reset -control_signal test_mode -test_data
rst_n

# 4. Create Test Protocol & Run Pre-Scan DRC
create_test_protocol
dft_drc -verbose > reports/dft_drc_pre_insert.rpt

# 5. Insert Scan Chains & Balance
preview_dft
insert_dft
```

10 Physical Design (PnR) Constraints & Technology Rules

PnR Floorplan, CTS & Routing Constraints

[Cadence Innovus / Synopsys ICC2]

Master Example:

```
# 1. Floorplan & Core Aspect Ratio Constraints
create_floorplan -core_utilization 0.60 \
                 -aspect_ratio 1.0 \
                 -core_margins_by die \
                 -core_margin_top 10.0 -core_margin_bottom 10.0 \
```

```

        -core_margin_left 10.0 -core_margin_right 10.0

# 2. Macro Placement Halo (Keepout Margins)
create_placement_halo -width 5.0 -sides {top bottom left right} [get_cells
    u_mem_macro]

# 3. Clock Tree Synthesis (CTS) Options
set_clock_tree_options -clocks [get_clocks SYS_CLK] \
    -target_skew 0.05 \
    -target_latency 0.80 \
    -max_transition 0.15 \
    -max_capacitance 0.10 \
    -buffer_relocation true

# 4. Non-Default Routing Rules (NDR: 2W2S for Clocks)
create_routing_rule RULE_2W2S -multiplier_width 2 -multiplier_spacing 2
set_routing_rule -rule RULE_2W2S [get_nets -of_objects [get_clocks *]]

# 5. Antenna Ratio Design Rules
set_antenna_rules -max_ratio 400.0 -diode_cell ANTENNA_X1

```

11 FPGA Constraints (Xilinx XDC & Intel Quartus QSF)

Xilinx Vivado XDC Constraints

[Physical Pinout & Timing]

Master Example:

```

# 1. Timing Clock Constraint
create_clock -period 10.000 -name sys_clk_pin -waveform {0.000 5.000} [
    get_ports clk]

# 2. Physical Pin Placement & IO Standards
set_property PACKAGE_PIN E3 [get_ports clk]
set_property IOSTANDARD LVCMOS33 [get_ports clk]

set_property PACKAGE_PIN C12 [get_ports rst_n]
set_property IOSTANDARD LVCMOS33 [get_ports rst_n]

set_property PACKAGE_PIN D4 [get_ports uart_tx]
set_property IOSTANDARD LVCMOS33 [get_ports uart_tx]
set_property DRIVE 8 [get_ports uart_tx]
set_property SLEW FAST [get_ports uart_tx]

# 3. Asynchronous Clock Domain Groups in XDC
set_clock_groups -asynchronous -group [get_clocks -include_generated_clocks
    sys_clk_pin] \
    -group [get_clocks -include_generated_clocks
    tx_clk_pin]

# 4. Max Delay Datapath Only for Asynchronous Crossings
set_max_delay -from [get_cells u_async_fifo/wr_ptr_reg*] \
    -to [get_cells u_async_fifo/sync_reg1*] 5.000 -datapath_only

```

12 Complete Master SDC Script for the Diploma System Chip

System_Top.sdc

[Unified Production SDC File]

```

#
#####

```

```

#                               MASTER PRODUCTION SDC CONSTRAINT FILE
#                               System Controller SoC
#

#####

current_design System_Top

# 1. Time Units & Operational Environment
set_operating_conditions -max scmetro_tsmc_cl013g_rvt_ss_1p08v_125c \
                        -max_library scmetro_tsmc_cl013g_rvt_ss_1p08v_125c

# 2. Clocks Definition
set CLK_PER 10.0    ;# 100 MHz Master System Clock
create_clock -name REF_CLK -period $CLK_PER [get_ports clk]
create_clock -name UART_CLK -period 80.0    [get_ports uart_clk]

# Virtual Clocks for I/O Timing Reference
create_clock -name VCLK_SYS -period $CLK_PER
create_clock -name VCLK_UART -period 80.0

# Generated Clocks
create_generated_clock -name TX_CLK -source [get_ports uart_clk] -divide_by 8 \
                        [get_pins u_clkdiv/clk_out]

create_generated_clock -name GATED_ALU_CLK -source [get_ports clk] \
                        -combinational \
                        [get_pins u_clk_gate/GATED_CLK]

# 3. Clock Uncertainties & Latencies
set_clock_uncertainty -setup 0.20 [get_clocks REF_CLK]
set_clock_uncertainty -hold 0.08 [get_clocks REF_CLK]
set_clock_uncertainty -setup 0.50 [get_clocks UART_CLK]
set_clock_uncertainty -hold 0.15 [get_clocks UART_CLK]

set_clock_transition 0.10 [all_clocks]

# 4. Input Constraints
set_input_delay -max [expr $CLK_PER * 0.25] -clock VCLK_SYS [get_ports rst_n]
set_input_delay -min [expr $CLK_PER * 0.05] -clock VCLK_SYS [get_ports rst_n]

set_input_delay -max 20.0 -clock VCLK_UART [get_ports uart_rx]
set_input_delay -min 4.0 -clock VCLK_UART [get_ports uart_rx]

set_driving_cell -lib_cell INV_X4 [all_inputs]
set_dont_touch_network [get_ports {clk uart_clk rst_n}]

# 5. Output Constraints
set_output_delay -max 20.0 -clock VCLK_UART [get_ports uart_tx]
set_output_delay -min -2.0 -clock VCLK_UART [get_ports uart_tx]
set_load 0.05 [all_outputs]

# 6. Timing Exceptions
set_clock_groups -asynchronous -group [get_clocks {REF_CLK GATED_ALU_CLK}] \
                -group [get_clocks {UART_CLK TX_CLK}]

set_false_path -from [get_ports rst_n]

set_max_delay 4.0 -datapath_only -from [get_cells u_async_fifo/ \
    wr_ptr_gray_reg*] \
                -to [get_cells u_async_fifo/u_sync/ \
                    sync_reg1*]

# 7. Design Rule Constraints (DRCs)

```

```
set_max_transition 0.20 [current_design]
set_max_capacitance 0.15 [current_design]
set_max_fanout 16 [current_design]
set_max_area 0

# 8. Path Optimization Groups
group_path -name INREG -from [all_inputs]
group_path -name REGOUT -to [all_outputs]
group_path -name INOUT -from [all_inputs] -to [all_outputs]
group_path -name CORE -critical_range 1.0 -weight 2.0 [get_clocks REF_CLK]
#
#####
```